Dataset & Research Objective: Forecasting DBS Group with a Hybrid Neuro-Fuzzy Model

Asset Tear-Sheet

DBS Group Holdings Ltd. D05.SI

Singapore-listed bank

Jan 2020 - Dec 2025

Raw Fields	Engineered Features
Open High Low Close Adj Close Volume	Daily Returns Moving Averages Rolling Volatility

Data Integrity Checks:
Missing observations, corporate actions, price adjustment consistency.

Training-set Normalization Only
(prevents information leakage)

30-Trading-Day Input Window

Predict Next-Day Close Price (T+1)

Preserve Chronological Split
(Train -> Val -> Test)

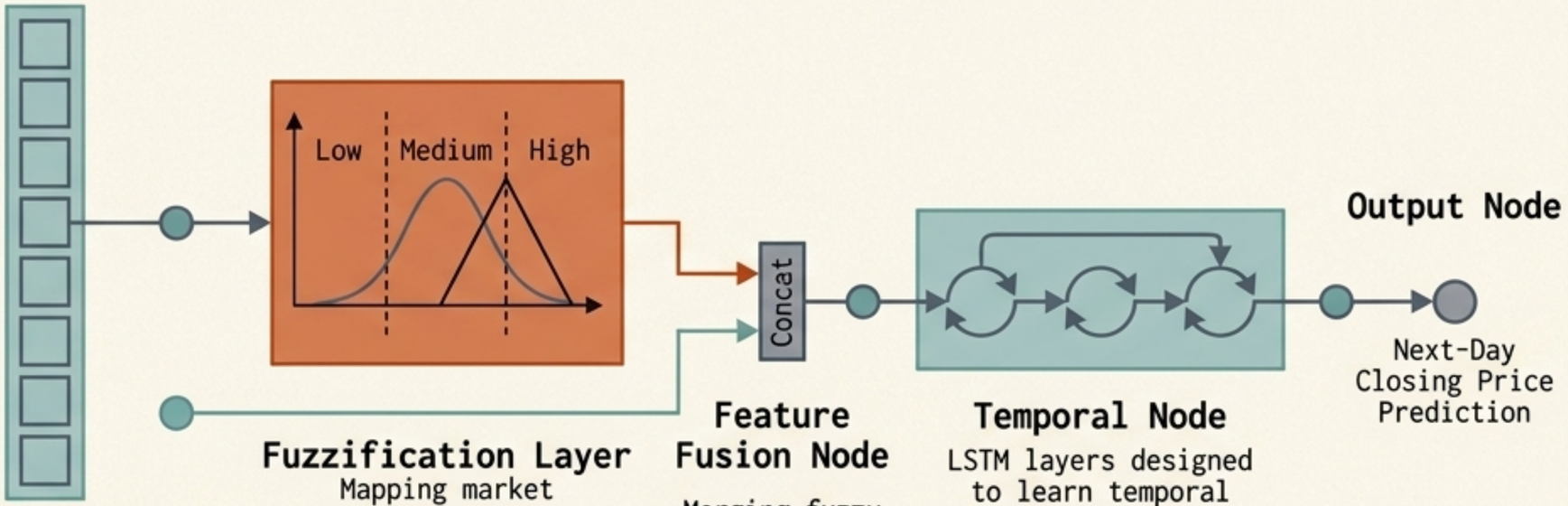
Project Hypotheses

H1: Does fuzzy representation improve LSTM price-forecasting accuracy?
H2: Can model forecasts produce better simulated trading returns under common rules?

DATA ACCESS: https://sg.finance.yahoo.com/quote/D05.SI/history/

Gemini Notebook

Hybrid AI Method: Fuzzy LSTM Architecture



Input Sequence
30-day sequences of historical prices & technical indicators

Fuzzification Layer
Mapping market features to membership functions

Feature Fusion Node
Merging fuzzy memberships and standard numeric inputs.

Temporal Node
LSTM layers designed to learn temporal patterns.

Output Node
Next-Day Closing Price Prediction

Implementation Protocol Matrix

Functions:

Triangular or Gaussian membership functions (ranges fitted exclusively on training data).

Optimization:

Trained via regression loss; hyperparameters tuned on validation data.

Ablation Study Plan:

Model A (Standard LSTM)	Model B (LSTM + Fuzzy Features)

PUBLIC CODE: https://github.com/MakhovaAnastasia/Paper_Fuzzy_LSTM
PUBLICATION DOI: https://doi.org/10.1007/978-3-032-15761-4_4

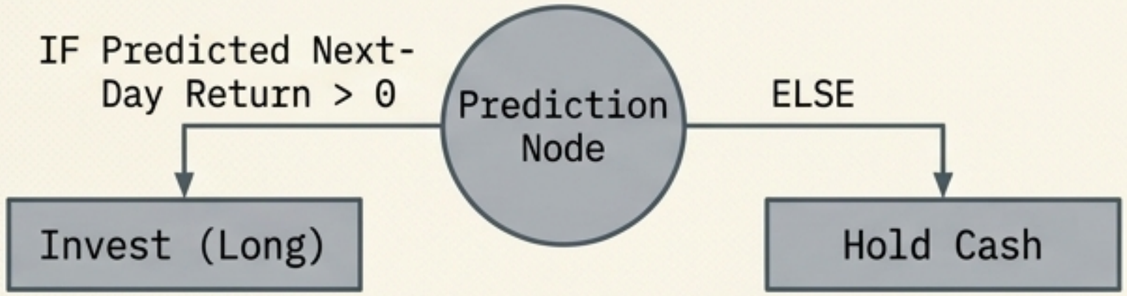
Gemini Notebook

Benchmark, Evaluation Methodology & Resources

Evaluation Framework Matrix

	Proposed Model (Fuzzy LSTM)	Baseline (Standard LSTM)
Data Alignment	Trained on identical target, 30-day windows, and DBS dataset	
Validation Strategy	Chronological holdout + Walk-forward evaluation	
Target Metrics	Primary: RMSE | Secondary: MAE	

Algorithmic Flow



Simulated trading returns evaluated strictly against a passive Buy-and-Hold strategy.

Deliverables:

- ☑ Reproducible
- ☑ Jupyter Notebook, literature review,
- ☑ results analysis, slide deck.

Guardrails:

Assessment of transaction costs and look-ahead bias (Improvements are hypotheses, not guaranteed).

Resource Directory

[1] DBS DATASET: https://sg.finance.yahoo.com/quote/D05.SI/history/
[2] FUZZY CODE: https://github.com/MakhovaAnastasia/Paper_Fuzzy_LSTM
[3] COLAB SAMPLE: https://colab.research.google.com/drive/14VEWf5KzSxrTm8V4Px6cDu3eN3KUJXk5

Gemini Notebook